Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral -Type 3 and Type 4

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral -Type 3 and Type 4

1 Particular integral -Type 3 and Type 4

Study focus: Particular integral -Type 3 and Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on finding the particular integral when x is a polynomial, that is x is x power m and.

Key details

- When x is epoch a x into v, where v is a function of x.
- As usual, in this case the particular integral is expressed in the form one over f of d times x power m.
- On all f of b is written as f of b inverse times x power m.
- Once we do this, we expand this in ascending power sub b using binomial series.
- Up to the term containing d power m it means the terms beyond d power m are neglected because.
- P power n of x power m is zero for n is greater than m.
- It means n is three and m is two.
- And you can clearly see n is better than m and it's going to be zero.
- When n is one, when you differentiate first time, x square become two x, seven x becomes seven.
- When n is two, two x becomes two, seven becomes zero, when n is three,.
- It becomes zero.
- It means the nth derivative of x power m is zero when n is m.
- So because of this condition, it comes beyond b power m or neglected.
- Solve the differential equation d square plus 3d plus 2 of y equals x power 3 plus x square.
- You can observe carefully the value of M is three here.
- The axillary equation is m square plus 3m plus 2 0.
- The roots are minus 1 and minus 2.

- plus c 2 times e power minus 2x.
- It is 1 over the d square plus 3d plus 2 of x square plus x square.
- from theory part, when we apply case number three, it is convenient to express f of d either in the form of 1 plus d or 1 minus d, so that we can apply binomial series.
- It is written as 2 of 1 plus d square plus 3d by 2.
- It becomes 1 plus 3d, 1 plus d square plus 3d by 2, something like minus 1.
- from logic whenever we have one plus it is alternative sign on the right hand side one minus x plus x square minus x power three and so on.
- And remember x is nothing but d square plus three d by two.

Example or application from the lecture

- Let us start with case number three when x is x power m.
- Let us take an example.
- Let us take x square plus seven x plus eight.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

This is x.

Key details

- This is x square.
- This is x power three and so on.
- Expanding this using binomial series up to the term containing d power m.
- Logically speaking, from x power 3 plus x square.
- The highest power of x is 3.
- It means when we use binomial series we should go only up to d power 3.
- The term beyond d power 3 are neglected, it means d power 4, d power 5, so on they're all neglected because anyway they become zero.
- Obviously this is considered, D square is considered, 3D is considered because they are less than 3.
- When you look at d square plus 3d whole square.
- We don't want a square.
- We want b square.
- We want 9 d square.
- And we don't want yeah we want 2ab that is 6d q.
- What about the fourth term?
- We don't want a power 3 but we want bq because it's 27 d power 3.
- If you have d of that, differentiate x power three plus x square once with respect to x.
- If it is d square, differentiate two time.
- If it is d power three, differentiate three times.
- The final PI is going to be half times x power 3 minus 7x square by 2 plus 30x by 4 minus 31 by 4.

- very well how to compute the complementary function.
- The roots are minus 1 plus r minus root phi by 2 into x.
- Then what about particular integral?
- Minus 25 by 9.
- Where b is a function of x.
- Remember, this is a combination of case four plus case three and case two.

Example or application from the lecture

- Let us look at another example. x square plus 2x minus 1.
- Again the task is to reduce this polynomial to one plus x mar let us take three common you get one plus two d square plus two d by three and this part looks like even x.
- Let us move on to case four.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

Depends on B.

Key details

- This is expressed in the form e po x times we choose a from here, replace a d by d plus c in f of b that is here.
- then retain v of x as it is.
- From here to here depends on case four it could be case two and it could be case three.
- If v is sign or cos, we go with case 2.
- If v of x is a polynomial, we go with case number 3.
- D square plus two of y equals x square times a power three x.
- Complementary function the roots are plus or minus root two i because alpha zero beta is root two the complementary function becomes C1 cos root two x plus c two sine root two x.
- You need to go by this logic.
- You have x square into e power 3x.
- If x square is not there, case 1.
- If e power 3x is not there, case 3.
- So now you should go by case number four because both are available.
- times e power 3x.
- This is written as a power 3x times 1 divided by.
- Replace d by d plus a.
- And you can see a values 3.
- and rate it be as it is and observe carefully because it's x square you go by case number three so when you simplify you get case plus 6 d is 11x square.
- It's a polynomial case.
- We want 1 plus x form.
- Then it is written as a power 3x by 11.
- Then apply binomial series we get 1 minus x plus x square minus x power 3.

- Because the highest power of x is 2, we need to go up to d square.
- This is considered.
- This is considered.
- A square is not needed.
- B square is needed.
- 2AB is not needed.
- And in this case, nothing is needed because it is more than 2.
- Then take x square inside.
- Then start up.
- Like differential operator, we get this.
- As usual, the complementary function is going to be c1 plus c2x times e power x because the roots are repeated one and one.
- And this is a particular integrity.
- This denominator equals d minus the whole square so problem becomes simple.
- So it means we need to apply inverse differential operator integrate sine x to times.

Example or application from the lecture

- Let us find out the solution to this differential equation.
- Let us move on to another problem where x is e power x times sine x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 15:00–19:59

Answer comes out to be minus a power x times sine x.

Key details

- When x is a polynomial, when x is epo x, epo a x into v, where v is a function of x.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

C Revision and retrieval practice